

Week 5 - RCBD

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Why use blocks?

- In CRD: randomisation of experimental units to treatments was completely unrestricted.
- All variables that might affect the response average out.
- So we are confident: differences in group means are due to treatments because possible differences between experimental units are spread out across the treatment groups.

Important sources of variability

- However, often there is important variation in additional variables that we are not directly interested in.
- Randomisation doesn't always guarantee perfect balance, especially when there is a strong source of variation.
- If we can group according to these variables to make experimental units more similar, we have a more powerful design.

Quick Example

- A researcher wants to test two different fertilizers (A and B) to see which one leads to the best plant growth.
- We can randomly assign fertilisers to pots in a greenhouse as we would in a CRD.
- But we know that one part of the greenhouse consistently receives more sunlight and sunlight is an important determinant of plant growth.
- Instead of relying on unrestricted randomisation, we can block to explicitly make experimental units within treatments more similar.
- Split the greenhouse into blocks based on the degree of sunlight and randomise within.

So when to use blocking?

- When there is a known source of variation that impacts the response.
- This variation is not of interest, but we want to control for it to get a clearer treatment effect.

Common blocking factors

- Geographic location: field, site, regions or cities that share similar economic conditions.
- Time: experimental replication over different days or weeks. Blocking for economic cycles or seasonal effects.
- Subject: person, plant, businesses, phenotype.
- Demographic groups: age, gender, income or education level, consumer behavior segments.
- Equipment: container types, growth chambers.

Example #1

A researcher is studying the effect of three different fertilizers (A, B, and C) on wheat yield. However, soil quality varies across the field, which could influence the results. To account for this, the field is divided into blocks based on soil type (sandy, loamy, and clayey). Within each block (soil type), the three fertilizers are randomly assigned to plots. The wheat yield (kg per plot) is recorded at the end of the growing season.

Example #2

A school is testing three different teaching methods (lecture-based, group discussion, and problem-solving approach) to see which leads to better student performance using different classrooms. The school has different teachers for sets of classrooms and they suspect that the unique characteristics of each teacher might influence the response. So they conduct a Randomised Complete Block Design with teacher as the blocking variable with the average grade of classrooms as the response.

Example #3

To compare the effects of three types of exercise (cycling, running, and rowing) on heart rate recovery, participants are first grouped based on fitness level (low, moderate, high). Within each fitness level, individuals are randomly assigned to one of the three exercises. After exercising, their heart rate recovery time (time taken to return to resting heart rate) is measured as the response.

SF RCBD

- We wish to compare a treatments and have N experimental units arranged in b blocks.
- Each block contains a homogeneous experimental units: $N = ab$.
- The a treatments are assigned to the units in the j^{th} block at random.
- This means that each treatment is replicated b times.

Example datasets #1

Wheat Yield (kg) Across Different
Fertilizer Treatments

Soil	A	B	C
Sandy	37.94	41.12	47.22
Loamy	39.78	45.23	50.35
Clayey	48.87	53.68	59.52

Example datasets #2

Test Scores Across Different Teaching Methods

Teacher	Lecture	Group_Discussion	Problem_Solving
Teacher 1	75	79	52
Teacher 2	80	58	94
Teacher 3	68	85	61
Teacher 4	90	73	88
Teacher 5	65	84	60

Example datasets #3

Heart Rate Recovery (seconds) Across Different Exercises

Fitness_Level	Cycling	Running	Rowing
Low	65	80	50
Moderate	60	78	45
High	55	72	42

The one-way ANOVA model

$$Y_{ij} = \mu + A_i + e_{ij}$$

where

$i = 1, \dots, a$ (a = number of treatments)

$j = 1, \dots, r$ (r = number of replicates)

Y_{ij} = response value of the j^{th} unit receiving treatment i

μ = overall or general mean

A_i = effect of the i^{th} level of treatment factor A

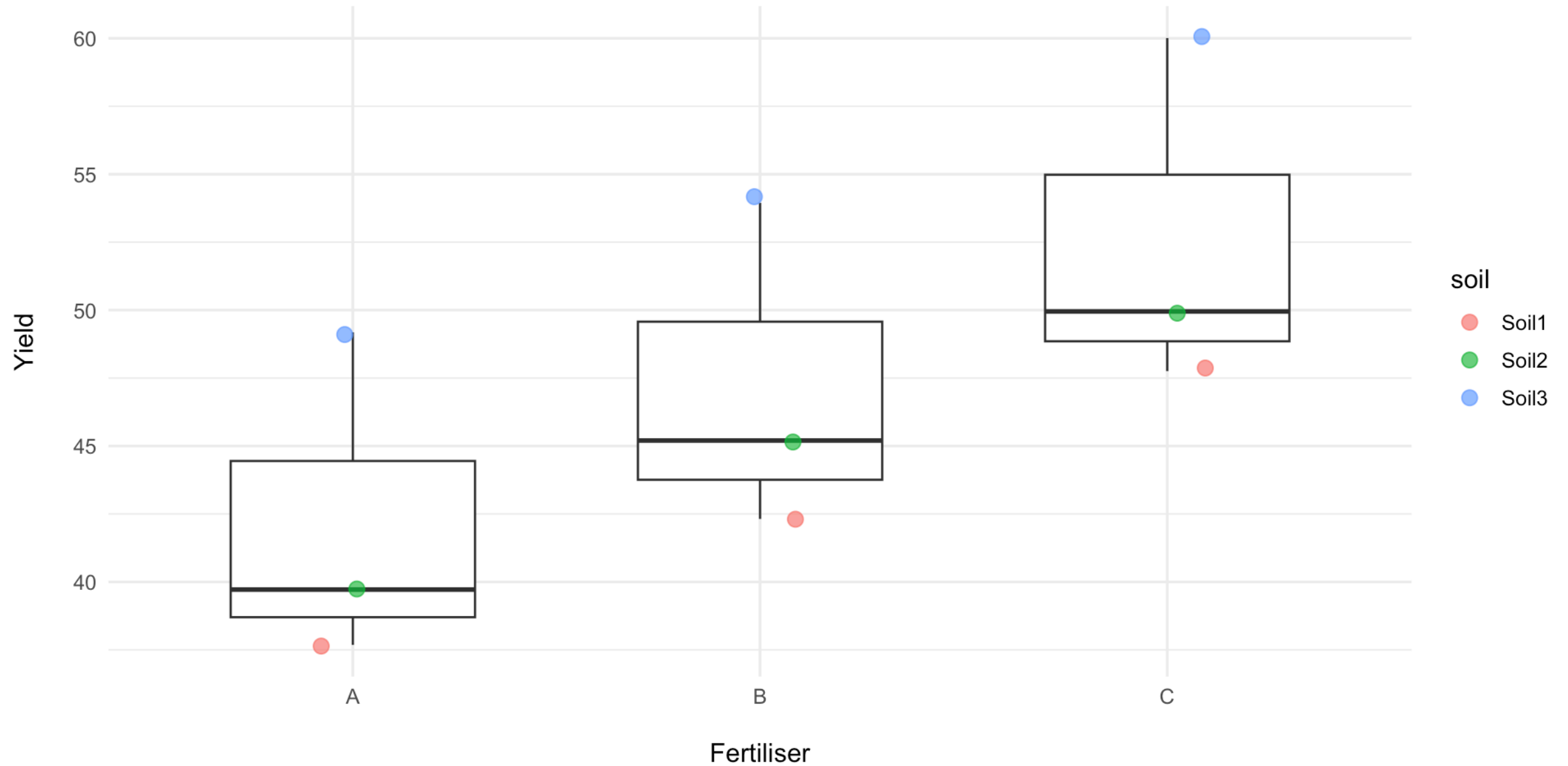
e_{ij} = random error with $e_{ij} \sim N(0, \sigma^2)$

Problem without blocking when necessary

If we ignore the blocking factor, some of the variation in e_{ij} might be due to systematic differences between blocks rather than pure randomness.

- Fertiliser Experiment: Soil type affects yield (i.e. induces variation in response) but if it is not included in the model, the error term will contain both random variation and the differences due to soil type.
- Teacher Experiment: If the “teacher” affects success but is not included in the model, the error term will contain both random variation and the teacher effect.
- Fitness Experiment: If fitness level affects heart rate but is not included in the model, the error term will contain both random variation and the fitness level effect.

Visualisation



Introducing a blocking term

To account for variation due to the blocking variable, we modify the model by adding a block effect:

$$Y_{ij} = \mu + A_i + B_j + e_{ij}$$

- B_j (block effect) represents systematic differences due to blocks (e.g., soil type, teacher, fitness level).
- The block effect is interpreted in the same way as the treatment effect, it is the difference between block mean and the overall mean.
- Estimation as before using least square estimators (i.e. sample means)

Sums of squares

Now we have an extra source of variability:

$$Y_{ij} = \mu + A_i + B_j + e_{ij}$$

$$Y_{ij} - \mu = (\mu_i - \mu) + (\mu_j - \mu) + (Y_{ij} - (\mu - A_i - B_j))$$

$$Y_{ij} - \bar{Y} = (\bar{Y}_i - \bar{Y}) + (\bar{Y}_j - \bar{Y}) + (Y_{ij} - (\bar{Y} - \hat{A}_i - \hat{B}_j))$$

$$\sum_i \sum_j (Y_{ij} - \bar{Y})^2 = \sum_i \sum_j (\bar{Y}_i - \bar{Y})^2 + \sum_i \sum_j (\bar{Y}_j - \bar{Y})^2 + \sum_i \sum_j (Y_{ij} - \bar{Y}_i)^2$$

$$SS_{\text{total}} = SS_{\text{treatments}} + SS_{\text{blocks}} + SS_{\text{within groups}}$$

Comparison to CRD

- CRD Decomposition:

$$SS_{\text{total}} = SS_{\text{treatments}} + SS_{\text{within groups}}$$

- RCBD Decomposition:

$$SS_{\text{total}} = SS_{\text{treatments}} + SS_{\text{blocks}} + SS_{\text{within groups}}$$

- If we didn't block and it was necessary:

$$SS_{\text{total}} = SS_{\text{treatments}} + [SS_{\text{blocks}} + SS_{\text{within groups}} = SS_{\text{within groups}}]$$

ANOVA Table

Source	SS	df	MS	F
Treatments A	$SS_A = b \sum_i (\bar{Y}_i - \bar{Y}_{..})^2$	$(a - 1)$	$\frac{SS_A}{(a-1)}$	$\frac{MS_A}{MSE}$
Blocks B	$SS_B = a \sum_j (\bar{Y}_j - \bar{Y}_{..})^2$	$(b - 1)$	$\frac{SS_B}{(b-1)}$	$\frac{MS_B}{MSE}$
Error	$SSE = \sum_{ij} (Y_{ij} - \bar{Y}_{i.} - \bar{Y}_{.j} + \bar{Y}_{..})^2$	$(a - 1)(b - 1)$	$\frac{SSE}{(a-1)(b-1)}$	
Total	$SS_{total} = \sum (Y_{ij} - \bar{Y}_{..})^2$	$ab - 1$		

In R

```
1 crd_model <- aov(Yield ~ Fertilizer, data = fertilizer_long)
2 summary(crd_model)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Fertilizer	2	155.8	77.88	2.012	0.214
Residuals	6	232.2	38.71		

```
1 rcdb_model <- aov(Yield ~ Fertilizer + Soil, data = fertilizer_long)
2 summary(rcdb_model)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Fertilizer	2	155.77	77.88	210.6	8.85e-05 ***
Soil	2	230.75	115.38	312.0	4.06e-05 ***
Residuals	4	1.48	0.37		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

ANOVA Hypotheses

Now we have two sets of hypotheses:

1. Treatments:

H0: treatment means are equal vs **H1: At least one treatment mean is different.**

2. Blocks:

H0: block means are equal vs **H1: At least one block mean is different.**

F-test for blocks

- If $F > 1$ then blocking did reduce unexplained error variance.
- If $F \approx 1$ then the blocks did not improve the power of the experiment and you would have been equally well off with a CRD.
- If $F < 1$ (rare), blocking increased variability, indicating poor block choice or strong block-treatment interactions.

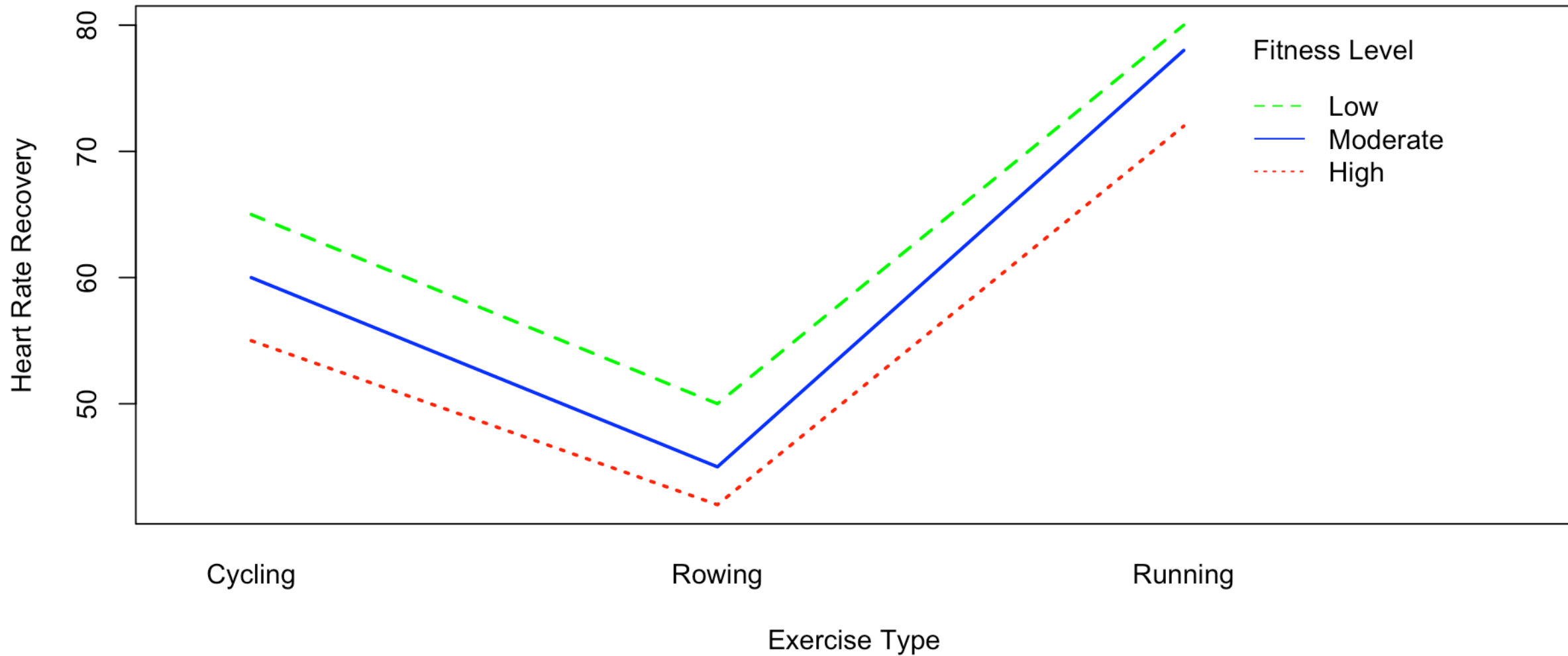
Analysis of RCBDs

- To analyse data from RCBDs we will use a two-way ANOVA without interactions
- Without interaction: we assume the treatment effects are similar in all blocks.

Additivity assumption

- This assumptions says that the treatment effect does not differ depending on the block.
- Teacher: the effect of each treatment is the same across teachers.
- Fertiliser: the effect of each fertiliser is the same across soil types.
- Fitness: the effect of each exercise type is the same across fitness levels.

Check in R



Assumptions of RCBD

Other assumptions are the same:

1. No severe outliers
2. Homogeneity of variance
3. Normality of errors
4. Independent errors

Apply to example

Heart Rate Recovery (seconds) Across Different Exercises

Fitness_Level	Cycling	Running	Rowing
Low	65	80	50
Moderate	60	78	45
High	55	72	42

In R

```
1 rcbd_model <- aov(Score ~ Exercise_Type + Fitness_Level,  
2                   data = exercise_long)
```

Model estimates

```
1 model.tables(rcbd_model, type = "means")
```

Tables of means

Grand mean

60.77778

Exercise_Type

Exercise_Type

Cycling Rowing Running

60.00 45.67 76.67

Fitness_Level

Fitness_Level

High Low Moderate

56.33 65.00 61.00

Model estimates

```
1 model.tables(rcbd_model, type = "effects")
```

Tables of effects

Exercise_Type		
Exercise_Type		
Cycling	Rowing	Running
-0.778	-15.111	15.889

Fitness_Level		
Fitness_Level		
High	Low	Moderate
-4.444	4.222	0.222

ANOVA in R

```
      Df Sum Sq Mean Sq F value    Pr(>F)
Exercise_Type  2 1444.2    722.1   649.9 9.41e-06 ***
Fitness_Level  2  112.9     56.4    50.8 0.00143 **
Residuals      4    4.4      1.1
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```